Alex Kipnis

PhD Researcher in Machine Learning

Simulation-Based Inference · Grouped Tabular Data · Benchmarking

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I work on amortized Bayesian inference, mixed-effects data and model evaluation: I built a PFN-style posterior estimator for mixed models (**MetaBeta**), a benchmark of 649 real grouped datasets (**MixedBench**) and an open data-generating process for them (**TabDGP**). Trained in machine learning, statistics and cognitive neuroscience.

RESEARCH EXPERIENCE

2020 – present

Doctoral Researcher, group of Eric Schulz

Helmholtz Munich (since 2024); MPI for Biological Cybernetics, Tübingen (2020–2023)

- **MetaBeta** (under review). PFN-style in-context posterior estimator for generalized linear mixed models, pretrained on millions of simulated grouped tables. It takes the model and its prior as test-time inputs. It matches NUTS in recovery and calibration, reproduces NUTS posteriors on **40 held-out real datasets** and stays indistinguishable from NUTS under misspecified likelihoods and priors, collinear and out-of-distribution designs and data-poor groups. MetaBeta takes only 0.1 s per dataset on one GPU (**500–1,400× faster than NUTS**; 90–210× on CPU).
  GitHub · PyPI · Colab · Hugging Face
- **MixedBench** (released on PyPI). A collection of documented mixed models of **649 public grouped datasets**, each fetched at a pinned version and checksum. Planned tracks: *posteriors* against brms references, and *prediction* on held-out rows and held-out groups with TabPFN (with and without the group id), GPBoost and glmmTMB as baselines.
  GitHub · PyPI · Colab
- **TabDGP** (alpha, released on PyPI). Data-generating process for grouped and flat tabular datasets, based on random structural causal models with **1–3 nested or crossed grouping factors** inside the graph. The outcome comes from the graph or from a mixed model with its own prior. Fidelity to real tables is scored by a random-forest two-sample test.
  GitHub · PyPI
- **MetaBench** (**ICLR 2025**). Compresses six Open LLM Leaderboard benchmarks to **under 3% of their size** using item response theory, recovering held-out models' scores at 1.24-point RMSE (0–100 scale). Included in EleutherAI's lm-evaluation-harness.
  GitHub · lm-eval-harness

2020

Staff Associate & Master's Thesis, Kriegeskorte Lab

Columbia University, New York

Validated statistical inference methods for representational similarity analysis with simulated fMRI data.

2019 – 2020

Research Intern & Research Assistant

MPI for Human Cognitive and Brain Sciences, Leipzig; Biological Psychology, LMU Munich

SELECTED PUBLICATIONS & PREPRINTS

2026 — Prior-Amortized In-Context Bayesian Inference for Generalized Linear Mixed-Effects Models. **Under review**, arXiv:2609.24422.
Alex Kipnis, Marcel Binz, and Eric Schulz.

2026 — Measuring what AI systems might do: Towards a measurement science in AI. **Minds and Machines**.
Konstantinos Voudouris, Mirko Thalmann, **Alex Kipnis**, José Hernández-Orallo, and Eric Schulz.

2025 — A sparse benchmark of reasoning and knowledge in Large Language Models. **ICLR 2025**.
Alex Kipnis, Konstantinos Voudouris, Luca M. Schulze Buschoff, and Eric Schulz.

2025 — A foundation model to predict and capture human cognition. **Nature**.
Marcel Binz, Elif Akata, …, **Alex Kipnis**, …, and Eric Schulz.

2023 — Statistical inference on representational geometries. **eLife**.
Heiko Schütt, **Alex Kipnis**, Jörn Diedrichsen, and Nikolaus Kriegeskorte.

EDUCATION

2020 – present — **Doctoral studies in Computer Science** (thesis submission expected May 2027)
University of Tübingen · Helmholtz Munich

2018 – 2020 — **M.Sc. Neuro-Cognitive Psychology**
LMU Munich · grade 0.9 · thesis at Columbia University

2015 – 2018 — **B.Sc. Psychology**
Heidelberg University · grade 1.2

2020 – 2023 — **Additional coursework in statistics, machine learning and computer science**
42 graduate ECTS in statistics and machine learning (LMU Munich, University of Tübingen; grade 1.1); 81 undergraduate ECTS in statistics and computer science (LMU Munich; grade 1.2).

TECHNICAL SKILLS

Programming — Python (PyTorch, NumPy, pandas, scikit-learn, PyMC); R (brms, lme4); Bash; Git.

Engineering — Test-driven, type-annotated research code; GitHub Actions CI and releases to PyPI, Zenodo and Hugging Face; training on SLURM clusters with Weights & Biases.

Methods — Amortized and simulation-based inference (normalizing flows, Set Transformers); MCMC (HMC/NUTS, Metropolis–Hastings); hierarchical Bayesian models and posterior evaluation (recovery, coverage, simulation-based calibration); structural causal models as simulators of tabular data; item response theory and factor analysis.

Open source — `metabeta`, `mixedbench`, `tabdgp` and `scamd` (Python), `metabench` (R); contributor to `bayesflow`, `rsatoolbox` and The Book of Statistical Proofs.

Languages — English, German, Russian.

SERVICE & AWARDS

2026 — Reviewer: NeurIPS 2026; ICLR 2026 RSI Workshop

2023 – 2026 — PhD scholarship, German Academic Scholarship Foundation (Studienstiftung)